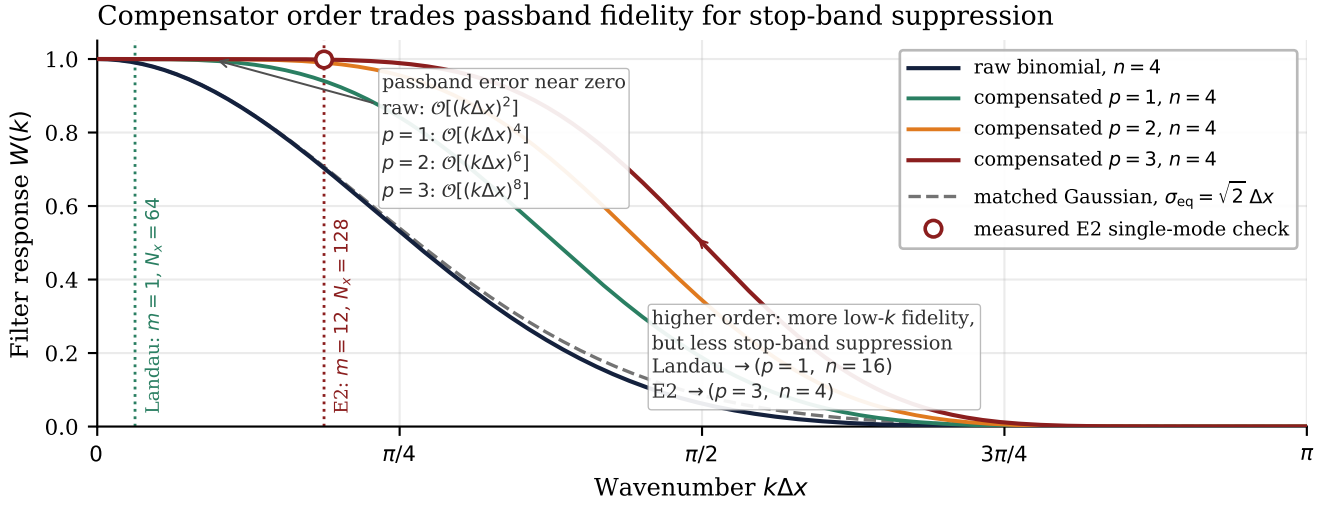
Compensator order trades passband fidelity for stop-band suppression
Filter response $W(k)$
Wavenumber $k\Delta x$
1.0
0.8
0.6
0.4
0.2
0.0
0
$\pi/4$
$\pi/2$
$3\pi/4$
π
raw binomial, $n = 4$
compensated $p = 1$, $n = 4$
compensated $p = 2$, $n = 4$
compensated $p = 3$, $n = 4$
matched Gaussian, $\sigma_{\mathrm{eq}} = \sqrt{2}\,\Delta x$
measured E2 single-mode check
passband error near zero
raw: $\mathcal{O}[(k\Delta x)^2]$
$p = 1$: $\mathcal{O}[(k\Delta x)^4]$
$p = 2$: $\mathcal{O}[(k\Delta x)^6]$
$p = 3$: $\mathcal{O}[(k\Delta x)^8]$
higher order: more low-k fidelity,
but less stop-band suppression
Landau $\rightarrow (p = 1,\ n = 16)$
E2 $\rightarrow (p = 3,\ n = 4)$
Landau: $m = 1$, $N_x = 64$
E2: $m = 12$, $N_x = 128$